# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 6 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The first term of a pattern is 9 and the 21st term is 169. The common difference is: 1

**(a)** 8 **(b)** 7 **(c)** 10 **(d)** 9

**2.** A pattern adds 1, then 2, then 3, ... Starting at 7, the term after 18 additions is: 1

**(a)** 179 **(b)** 177 **(c)** 178 **(d)** 196

**3.** The sum $1^3 + 2^3 + \cdots + 11^3$ equals: 1

**(a)** 4355 **(b)** 4357 **(c)** 4422 **(d)** 4356

**4.** The 68th triangular number minus the 67th triangular number is: 1

**(a)** 69 **(b)** 67 **(c)** 70 **(d)** 68

**5.** In a party 25 people greet each other once with a hug. Number of hugs: 1

**(a)** 300 **(b)** 301 **(c)** 275 **(d)** 325

**6.** The sum $1 + 2 + 3 + \cdots + n$ equals 2016. Then $n$ is: 1

**(a)** 63 **(b)** 65 **(c)** 62 **(d)** 64

**7.** The sum of the first 50 odd numbers is 2500. The sum of the odd numbers from 41 to 99 is: 1

**(a)** 2120 **(b)** 2101 **(c)** 2100 **(d)** 2099

**8.** The difference between the squares of the 35th and 34th numbers is: 1

**(a)** 71 **(b)** 70 **(c)** 68 **(d)** 69

**9.** The first term of a pattern is 4 and the 12th term is 169. The common difference is: 1

**(a)** 15 **(b)** 14 **(c)** 16 **(d)** 17

**10.** A pattern adds 1, then 2, then 3, ... Starting at 40, the term after 20 additions is: 1

**(a)** 270 **(b)** 251 **(c)** 250 **(d)** 249

## SECTION — B (Very Short Answer Type) 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Find the sum of odd numbers from 27 to 65. 2

**12.** 19 cricket teams play a league in which every team plays every other team once. Find the number of matches. 2

**13.** The sum of the first $n$ odd numbers is 3025. Find $n$ and the 55th odd number. 2

**14.** Find the sum of odd numbers from 33 to 81. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** In a pattern the 1st term is 17, the 2nd is 25, and so on by adding 8. Which term of the pattern equals 425? Find also the term just before it. **3**

**16.** Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 79 and use it to find $40^2$. Then evaluate the sum from 81 to 99. **3**

**17.** A tower is built with rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 10, and in the first 10 rows together? **3**

**OR**

In a pattern the 1st term is 4, the 2nd is 18, and so on by adding 14. Which term of the pattern equals 494? Find also the term just before it. **3**

**18.** Sums of consecutive odd numbers give squares. Find the sum of the odd numbers from 1 to 89 and use it to find $45^2$. Then evaluate the sum from 91 to 107. **3**

**OR**

A tower is built with rows of cubes: row 1 has 1 cube, row 2 has 3, row 3 has 6, and so on (triangular numbers). How many cubes are in row 12, and in the first 12 rows together? **3**

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** Look at the pattern: the 7th figure has 39 dots. How many dots are in figure 19? If a figure has 124 dots, which figure is it? Show that the dots form a pattern with common difference 5. **5**

**OR**

Two sequences: A is $25, 28, 31, 34, \ldots$ and B is $4, 11, 18, 25, \ldots$. Find the 20th term of each and decide which is larger and by how much. **5**

**20.** Look at the pattern: the 9th figure has 70 dots. How many dots are in figure 15? If a figure has 158 dots, which figure is it? Show that the dots form a pattern with common difference 8. **5**

**OR**

Two sequences: A is $9, 18, 27, 36, \ldots$ and B is $30, 35, 40, 45, \ldots$. Find the 13th term of each and decide which is larger and by how much. **5**

**END OF QUESTION PAPER** — **Total: 40 Marks**

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